AniMask: Anime-Informed Masking for MAE

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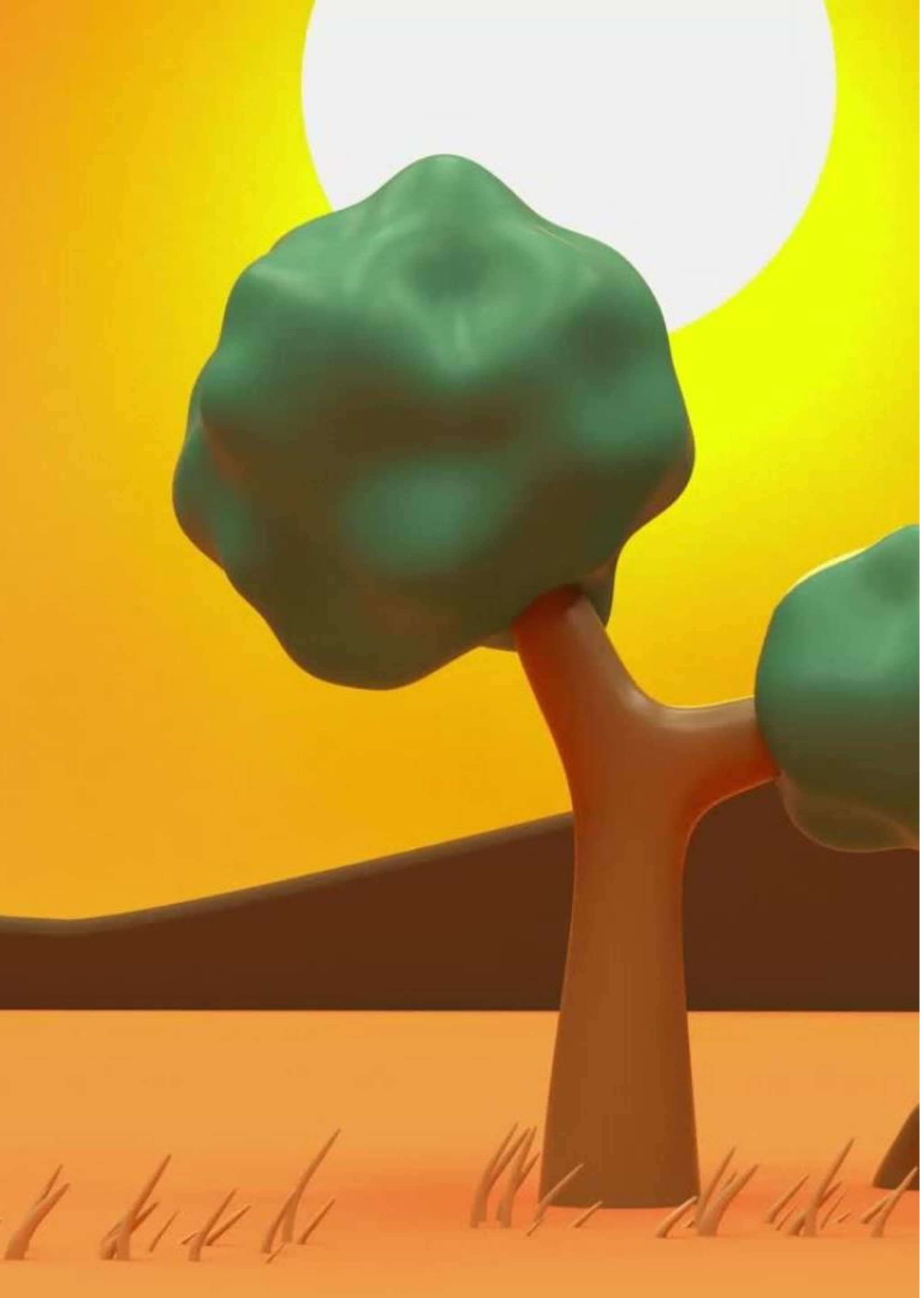

Motivation

- Most of researches only focus on natural images, rarely have researches only focus on anime images.
- Unlike natural images, anime imagery features highly structured visual hierarchies with distinct semantic elements such as character faces, expressive eyes, and clear foreground-background separation.
- Also, high intra-class variability across different art styles and artists .
- We want to find out if classical computer vision researches and their methods still works for anime images with high saturation, contrast, and sharp edges.

Comparison between natural images and anime images

PICS FROM HTTPS://IMGUR.COM/A/AFISV

Comparison between natural images and anime images

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Introduction

- Our project investigates semantic-aware masking strategies for Masked Autoencoders[1](MAE) specifically tailored to anime artwork.
- Although the authors of MAE mentioned that this method only uses small dataset training for a short period of time for realistic images, they still used more than 100k images and trained almost two days on 8 GPUs.
- We want to figure out if we can use a much smaller dataset with affordable GPUs on personal computers to have similar results, only focusing on anime images.

[1]:https://arxiv.org/abs/2111.06377

Research Goals
Our work addresses four fundamental questions:

- 1. Do ImageNet-pretrained features transfer effectively to anime imagery?
- 2. How much performance can be recovered through parameter-efficient adaptation？
- 3. Does domain-specific pretraining (anime→anime) outperform cross-domain transfer (ImageNet→anime)?
- 4. What is the quantitative value of pretraining compared to training from scratch?

Experimental Setup

- Hardware:
- - GPU: NVIDIA RTX 5090 (32GB VRAM)
- - CPU: Intel Core i9-14900K
- - RAM: 64GB DDR5
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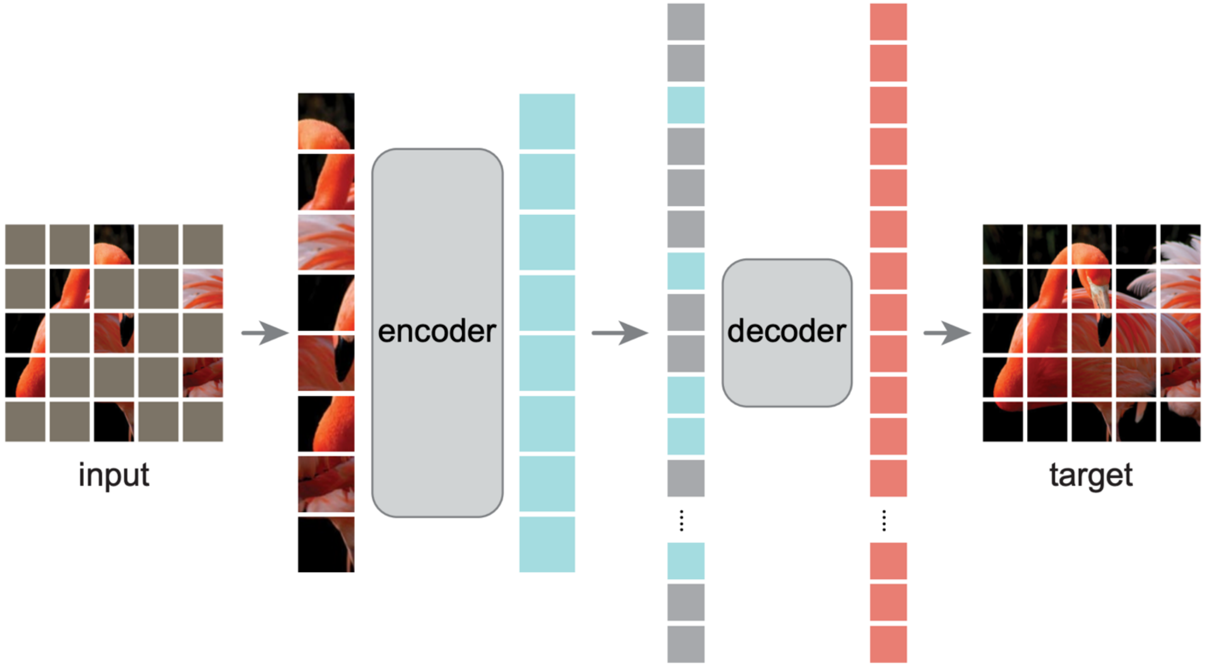
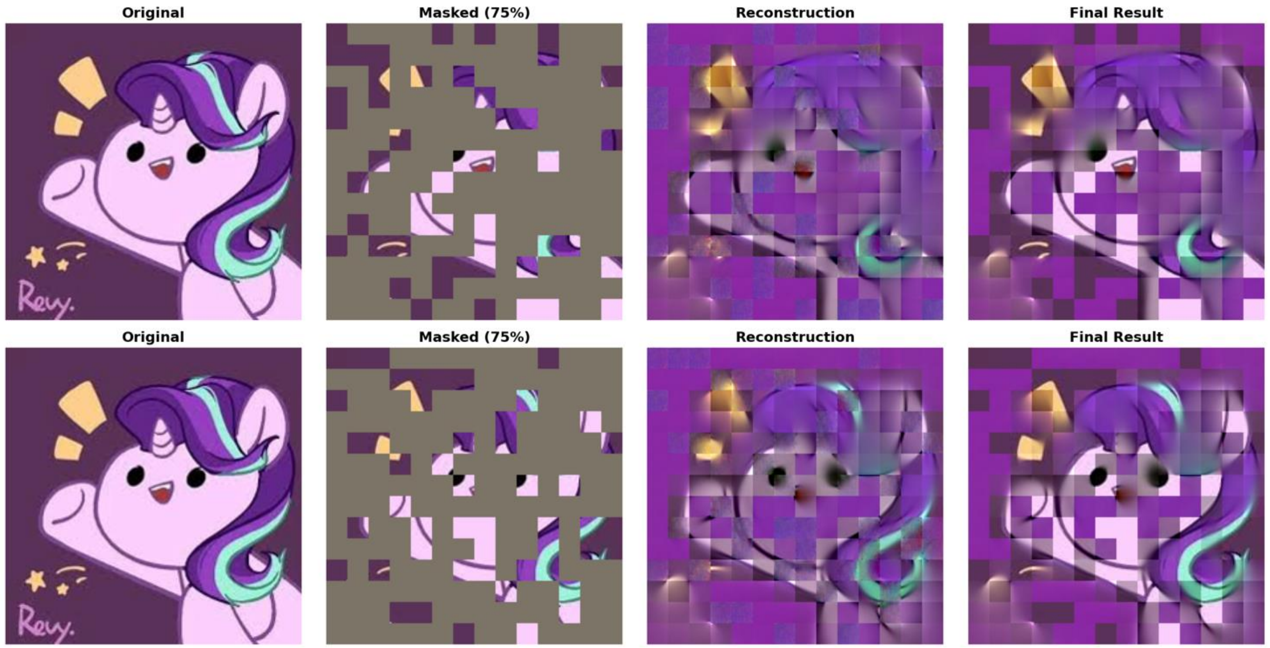
- Software:
- - Python: 3.10
- - PyTorch: 2.1.0
- - CUDA: 12.1
- - timm: 0.9.12

Baseline Experiment: Training from Scratch Subtitle: *Evaluation on Anime Diffusion Dataset*



Experimental Setup:Dataset: Trained on Anime Diffusion Dataset (High-quality character images).
Method: Trained ViT-Base from scratch (random initialization) without ImageNet pretraining.
Configuration: 75% Masking Ratio


Original
Masked (75%)
Reconstruction
Final Result
Original
Masked (75%)
Reconstruction
Final Result

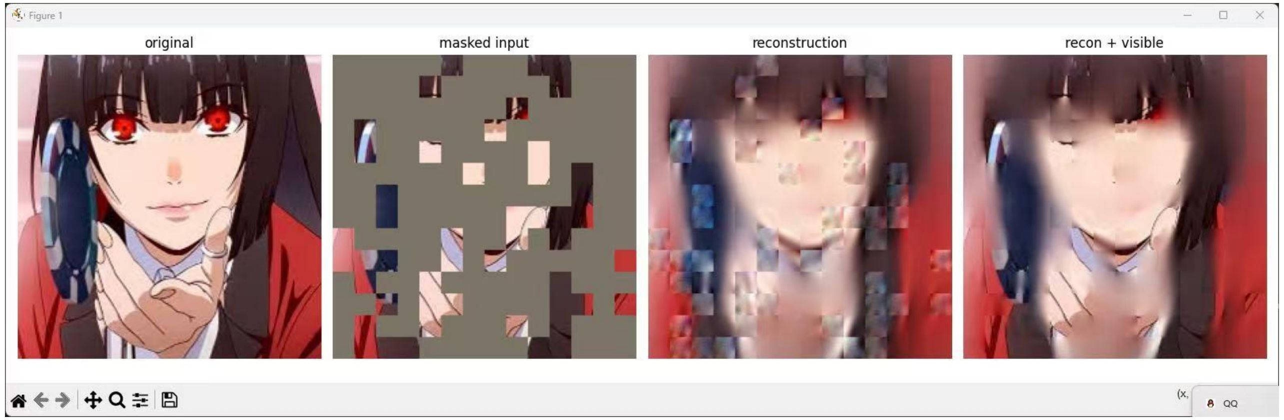


Figure 1
original
masked input
reconstruction
recon + visible

Experimental Dataset

- AnimeFace Character Dataset:
- - Source: Kaggle (https://www.kaggle.com/datasets/splcher/animefacedataset)
- - Total classes: 130
- - Total images: 12,853
- - Train/Val split: 9,340 / 3,513 (72.6% / 27.4%)
- - Average images per class: 98.8
- - Min images per class: 20
- - Max images per class: 250
- - Image resolution: Variable (resized to 224×224)

Example images of the dataset

Experiment Design

We conduct four experiments to systematically study transfer learning:

- Experiment 1: Training from Scratch
- **Configuration:** We initialized the ViT-Base model with **random weights** and trained all parameters end-to-end for 100 epochs.
- **Purpose:** To establish a **no-pretraining baseline**. We wanted to prove that our small dataset (12k images) is insufficient to train a Vision Transformer from scratch, which justifies the need for transfer learning.
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- Experiment 2: ImageNet Linear Probing
- **Configuration:** We loaded the official **ImageNet-pretrained weights** and **froze the entire encoder**. We only trained the final classification head (approx. 100k parameters).
- **Purpose:** To measure the **domain gap**. This tests how well features learned from natural images (like photos of objects) transfer to stylized anime art without any adaptation.

- Experiment 3: ImageNet LoRA Fine-tuning
- **Configuration:** We used the ImageNet-pretrained encoder but injected **trainable LoRA adapters** (Rank=8) into the attention and MLP layers. We kept the backbone frozen and only updated **1.13% of the parameters**.
- **Purpose:** To test **parameter-efficient domain adaptation**. We wanted to see if we could bridge the domain gap efficiently without the computational cost of full fine-tuning.
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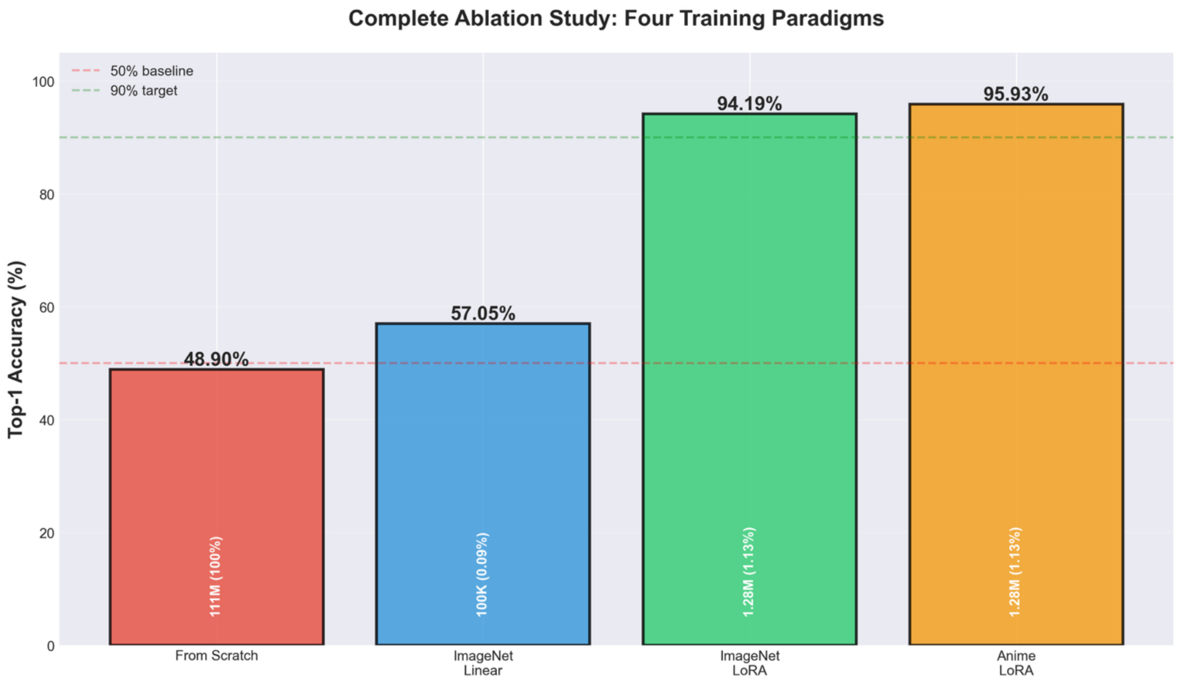
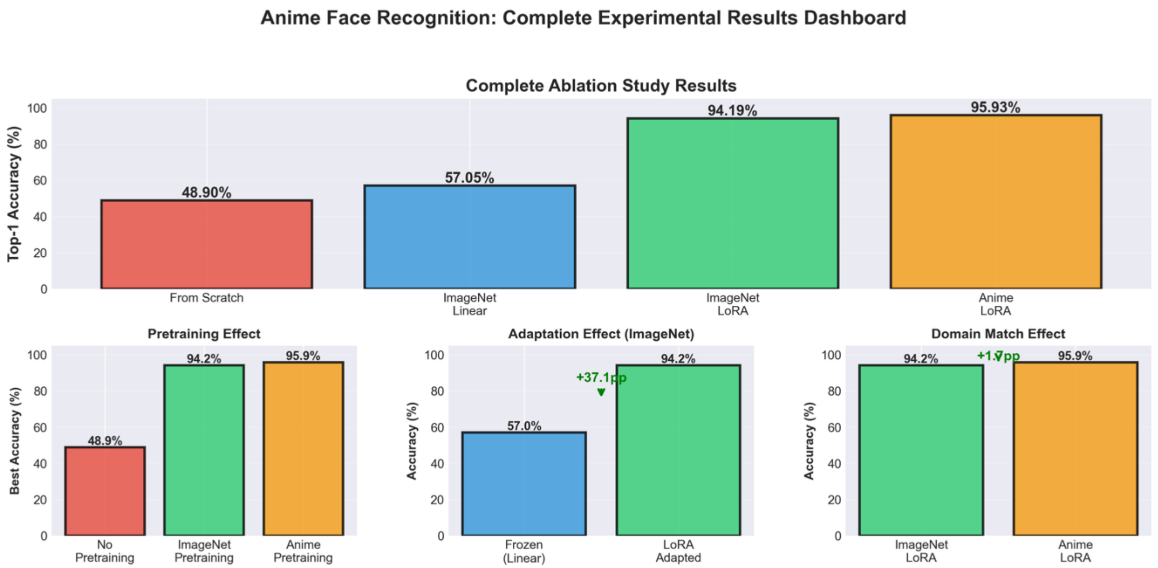
- Experiment 4: Anime LoRA Fine-tuning
- **Configuration:** Instead of ImageNet weights, we initialized the model with our own **domain-specific Anime-pretrained weights**. We then applied the same LoRA fine-tuning strategy.
- **Purpose:** To compare **in-domain vs. cross-domain pretraining**. This tests if starting with "anime knowledge" yields better results than starting with "natural image knowledge."

Experiment 3 and 4 Results





Method	Pretraining	Params	Top-1 Acc	vs Scratch	vs Linear
From Scratch	None	111M (100%)	48.90%	baseline	-
ImageNet Linear	ImageNet	100K (0.09%)	57.05%	+8.15pp	baseline
ImageNet LoRA	ImageNet	1.28M (1.13%)	94.19%	+45.29pp	+37.14pp
Anime LoRA	Anime	1.28M (1.13%)	95.93%	+47.03pp	+38.88pp

Findings:
The Domain Gap: Natural Images vs. Anime

- Our experiments reveal a substantial domain gap between ImageNet (natural photos) and anime (stylized artwork). Because while low-level edge features transfer, mid-to-high-level features (textures, lighting) require domain adaptation.

Findings:
Why LoRA Works: Decomposing the Success

- Low-Rank Bottleneck as Regularization
- Preserving Pretrained Knowledge
- End-to-End Feature-Label Co-Adaptation

Findings:
In-Domain vs. Cross-Domain Pretraining

- Anime LoRA (95.93%) outperforms ImageNet LoRA (94.19%) by +1.74 pp

- Domain-specific pretraining is preferable if computational resources allow
- +1.74 pp improvement** validates the hypothesis that domain match > data scale.
- However, ImageNet pretraining is still highly effective (94.19%), making it a strong fallback

Discussion:
From scratch achieves only 48.90% – why so poor?

1. Insufficient data:12K images far too small for training 111M parameters
2. Overfitting: Model likely memorizes training set without generalization
3. Random initialization: No low-level feature knowledge (edges, textures)
4. Limited capacity utilization: Unable to discover effective representations from scratch

Discussion:
Practical Recommendations

- 1. Use pretrained MAE (ImageNet or domain-specific)
- 2. Adapt with LoRA ($r=8$, $\alpha=16$) for parameter efficiency
- 3. Avoid training from scratch on limited data
- 4. Avoid linear probing as final solution

Limitations:

1. Single dataset: Results on AnimeFace (130 classes) may not generalize to all anime recognition tasks (e.g., different art styles, extreme occlusion, very low-resolution)

5. Computational constraints: Full fine-tuning experiment not conducted; ~96-97% upper bound is estimated from literature

Thank you!

References:

- [1] He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). **Masked Autoencoders Are Scalable Vision Learners.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 16000–16009.
- [2] Hu, E. J., Shen, Y., Wallis, P., et al. (2021). **LoRA: Low-Rank Adaptation of Large Language Models.** *arXiv:2106.09685*.
- [3] Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). **ArcFace: Additive Angular Margin Loss for Deep Face Recognition.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 4690–4699.
- [4] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). **A Simple Framework for Contrastive Learning of Visual Representations.** *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 1597–1607.
- [5] Jin, Y., & Takiguchi, T. (2018). **Anime Character Identification Using Multi-Task Learning.** *2018 25th IEEE International Conference on Image Processing (ICIP)*, 2268–2272.